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XIAMEN
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DIGITAL LOGIC

Assignment of Chapter One

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1). Fill-in-the-blank Questions

1. Digital circuits can only process digital signals, but they cannot process analog signals.
2. The machine code consists of sign-magnitude code , 1s complement code , and 2s complement code such three types of codes.
3. A binary number, 1001 1001, its decimal number is (153)₁₀ , its hex number is (99)₁₆ .
4. A binary number, 111 0101, its decimal number is (117)₁₀ , its octal number is (165)₈ .
5. A binary number, 0.110101, its octal number is (0.65)₈ , its hex number is (0.D4)₁₆ .
6. $[22.2]_8 = [10\ 010.01]_2$,
 $[18.A]_{16} = [11000.1010]_2$,
 $[63]_{16} = [143]_8 = [01100011]_2$.

- A decimal number 1102, its binary number is (100 0100 1110)₂, its octal number is (2116)₈, its hex number is (44E)₁₆.
- 8. A binary number -0.1011, its 1s complement code is (1.0100)_{1s}, its 2s complement code is (1.0101)_{2s}.
- 9. A binary number -10010, its sign-magnitude code is (11 0010)_{sign}, its 2s complement code is (10 1110)_{2s}.
- 10. A binary number -0.001101, its 1s complement code is (1.110010)_{1s}, its 2s complement code is (1.110011)_{2s}.
- 11. An 8421 code number, 0101 0110, its decimal number is (56)₁₀.
- 12. A 2421 code number, 1100 1110, its decimal number is (68)₁₀.

- 13. When a digital system processes analog signals, a/an **A/D D/A** device is used to convert the signals.
- 14. The integer's sign-magnitude number has **two** (one/two) representation(s) of "0".
- 15. The integer's 2s complement number has **one** (one/two) representation(s) of "0".

Short Answer Questions

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- 1 How to convert a Gray code number to a binary number?

Gray code

$$G = G_{n-1}G_{n-2} \dots G_{i+1}G_i \dots G_1G_0$$

Objective Binary code

$$B = B_{n-1}B_{n-2} \dots B_{i+1}B_i \dots B_1B_0$$

we have:

$$\left\{ \begin{array}{l} B_{n-1} = G_{n-1} \\ B_i = B_{i+1} \oplus G_i \end{array} \right. \quad 0 \leq i \leq n-2$$

□ 2How to add error correction to a parity code system?

Take the parity code check in the both vertical and horizontal directions:

1	1	0	0	1	0	1	1	→	1
0	1	0	1	1	1	0	0	→	0
1	0	0	1	1	0	1	0	→	0
1	0	0	1	0	1	0	1	→	0
↓	↓	↓	↓	↓					
1	0	0	1	1	0	0	0		

- 3 Convert the following binary number to the decimal, octal, and hex numbers, respectively.

- (1) 1110101 (2) 0.1110101 (3) 10111.01

$$\begin{aligned}
 (1) \quad (1110101)_2 &= 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + \\
 &\quad 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\
 &= 64 + 32 + 16 + 0 + 4 + 0 + 1 \\
 &= (117)_{10} = (165)_8 = (75)_{16}
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad (0.1110101)_2 &= 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} + 0 \times 2^{-4} \\
 &\quad + 1 \times 2^{-5} + 0 \times 2^{-6} + 1 \times 2^{-7} \\
 &= 0.5 + 0.25 + 0.125 + 0 + 0.03125 + 0 + 0.0078125 \\
 &= (0.9140625)_{10} = (0.724)_8 = (EA)_{16}
 \end{aligned}$$

$$\begin{aligned}
 (3) \quad (10111.01)_2 &= 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + \\
 &\quad 0 \times 2^{-1} + 1 \times 2^{-2} \\
 &= 16 + 4 + 2 + 1 + 0 + 0.25 \\
 &= (23.25)_{10} = (27.2)_8 = (17.4)_{16}
 \end{aligned}$$

- 4 Convert the following decimal numbers to the binary, octal, and hex number, respectively.
- (1) 29 (2) 0.27 (3) 33.33

$$(1) \quad 29 = (11101)_2 = (65)_8 = (1D)_{16}$$

$$(2) \quad 0.27 = (0.0100)_2 = (0.2)_8 = (0.4)_{16}$$

$$\begin{aligned} (3) \quad 33.33 &= (100001.0101)_2 \\ &= (41.24)_8 \\ &= (21.5)_{16} \end{aligned}$$

- 5 Try to use 8421 code and Gray code to represent the following binary number:
- (1) $(111110)_2$ (2) $(1100110)_2$

$$\begin{aligned}
 (1) \quad (111110)_2 &= (62)_{10} \\
 &= (0110 \ 0010)_{8421} \\
 &= (100001)_{\text{Gray}}
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad (1100110)_2 &= (102)_{10} \\
 &= (0001 \ 0000 \ 0010)_{8421} \\
 &= (1010101)_{\text{Gray}}
 \end{aligned}$$